<table>
<tr>
<td rowspan="2">Key partners
Nabeul municipality (host, owner)
Civil Protection (ONPC)
Regional committee (decides public alerts)
INM; Oued Souhil experimental station (data)
Ministry of Equipment (bridge access)
ANF (radio), INPDP (data protection)
Community associations (volunteers)
University lab (test bench)
Local electronics suppliers</td>
<td>Key activities
Site survey and radio planning
Building and installing nodes
Software: risk and escalation engines
Training technicians and volunteers
Drills with Civil Protection
Season report and threshold review</td>
<td rowspan="2">Value propositions
For people at the crossing: a beacon and siren that warn even with no network or phone.
For the municipality and ONPC: every alert confirmed by a person, automatic escalation when nobody answers, and a log of who did what.
For the regional committee: signed evidence (CAP) for its decision; humans keep authority.
For everyone: low cost, repaired locally, no vendor lock-in.</td>
<td>Customer relationships
Co-design of sites and messages
Joint drills; yearly training
Service contract with support
Public season report after each rainy season</td>
<td rowspan="2">Customer segments
Payers: municipalities; road authority (Ministry of Equipment); employers near wadis (sponsors); disaster-risk programmes (first sites)
Users: municipal duty officer, Civil Protection, volunteers
Beneficiaries: drivers, pedestrians, residents and workers near the crossing</td>
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<tr>
<td>Key resources
Open-source firmware and gateway software
One shared node design
Validated alert texts (AR/FR)
Trained volunteers and technicians
Repair manual; event log data</td>
<td>Channels
Direct meetings with municipalities and ONPC
Disaster-risk programmes and calls for projects
Pilot results and the open repository
Alerts: beacon, siren, SMS, voice, mesh radios</td>
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<tr>
<td colspan="3">Cost structure
Hardware and installation: about 10,500 TND per crossing (Table 4.3)
Running cost: about 760 TND per year (Table 4.4), plus support time
Fixed costs: part-time staff, travel, legal and certification
Free for the people protected: no cost to households or volunteers</td>
<td colspan="2">Revenue streams
Turnkey deployment: 13,500 TND per crossing
Optional service contract: 1,800 TND per site per year
Grants from disaster-risk programmes for pilots
Sponsorship of a crossing by a nearby employer
Alerts are always free for the public.</td>
</tr>
</table>

Figure 1: Business Model Canvas for NEREUS. Prices are planning assumptions (Section 4.4).